

Automated Game State Reconstruction: A Comprehensive Technical Framework for AI-Driven Football Analytics

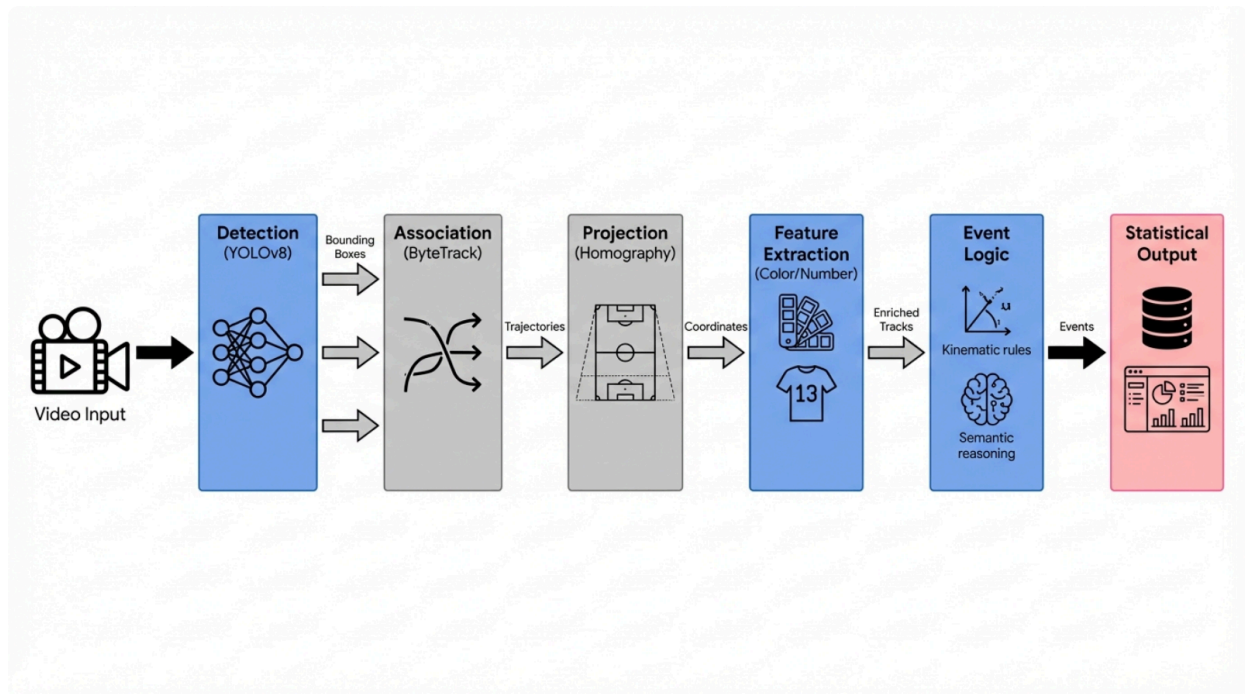
1. Introduction: The Paradigm Shift in Football Intelligence

The domain of football analytics is currently undergoing a radical transformation, driven by the migration from manual notation to automated, computer vision-based systems. Historically, the extraction of match data was a laborious process reliant on human annotators who would manually tag events—passes, shots, tackles, and fouls—from video feeds. This approach, while establishing the foundational datasets for the industry, suffered from inherent limitations: subjectivity in event definitions, low temporal resolution, and substantial latency between match completion and data availability.¹

Today, the convergence of deep learning architectures has enabled the automated extraction of spatiotemporal data from single-camera broadcast feeds with increasing fidelity. This report delineates the end-to-end architecture of a modern AI football analysis system designed to detect entities (ball, players, goalkeepers, referees), analyze semantic attributes (team affiliation, jersey numbers), and derive advanced statistical metrics. The system leverages state-of-the-art models such as YOLOv8 for object detection, ByteTrack for multi-object tracking, and Large Vision-Language Models (LVLMs) like Qwen2.5-VL for complex semantic reasoning.¹

The primary objective of such a system is Game State Reconstruction (GSR): the conversion of unstructured raw video pixels into a structured, queryable database of player trajectories and game events. This process is critical not only for elite professional clubs seeking marginal gains but also for the democratization of analytics, allowing semi-professional and amateur teams to access tracking data using consumer-grade video equipment.²

Architectural Data Flow: From Pixels to Performance Metrics



The processing pipeline begins with raw video ingestion, passing through parallel detection streams for players, ball, and referees. Trajectories are stabilized via ByteTrack and mapped to a 2D pitch model using homography. Semantic layers (Team ID, Jersey OCR) enrich the tracks before the Event Detection Engine applies kinematic rules to generate statistical outputs.

The following sections provide an exhaustive technical breakdown of this pipeline, exploring the algorithmic nuances of computer vision, the heuristics of event detection, and the statistical rigor required to generate metrics such as Expected Goals (xG) and Packing.

2. The Computer Vision Pipeline: Detection and Tracking Architecture

The foundational layer of any football analytics system is the accurate extraction of player and ball coordinates from video frames. This process, known as optical tracking, converts unstructured video data into structured tabular data containing (x, y) coordinates for every entity on the pitch at every timestamp (typically 25 or 30 frames per second). The robustness of this layer dictates the quality of all downstream analytics; errors in detection propagate into tracking failures, which in turn corrupt event detection logic.

2.1 Object Detection with YOLOv8

The industry standard for real-time object detection in sports is the YOLO (You Only Look

Once) architecture, specifically the YOLOv8 iteration. Unlike two-stage detectors such as Faster R-CNN, which generate region proposals before classifying them, YOLOv8 processes the entire image in a single forward pass. It divides the input frame into a grid and predicts bounding boxes and class probabilities for each grid cell simultaneously.⁵ This architecture is particularly well-suited for sports analytics due to its balance of inference speed and detection accuracy, essential for processing high-definition video feeds where player velocity can change rapidly.

In the context of football, the detection model must be fine-tuned to recognize four distinct classes, each presenting unique challenges:

1. **Players:** The primary subjects of analysis. The model must generalize across diverse kits, lighting conditions, and poses (e.g., sliding, jumping).
2. **Goalkeepers:** Goalkeepers are distinct from outfield players in both rules (handling the ball) and movement patterns. Their detection often relies on differentiating their jersey colors from the rest of the team or their relative positioning within the penalty area.⁵ Accurate classification of the goalkeeper is vital for metrics involving shot-stopping and defensive organization.
3. **Referees:** Referees must be explicitly detected and excluded from tactical analysis to prevent data contamination. Including a referee in a "defensive line" calculation would skew offside lines and packing metrics.
4. **The Ball:** Detecting the football is the most technically demanding task. The ball typically occupies less than 10 pixels in wide broadcast shots, moves at velocities exceeding 100 km/h, and is frequently occluded by players or merged with background clutter (e.g., white lines, advertising boards).⁶

Standard pre-trained models (like those trained on the COCO dataset) are insufficient for this domain because they lack the granular distinction between a "person" (which includes players, referees, and coaches) and the specific classes required for tactical analysis. Therefore, custom fine-tuning on annotated football datasets is mandatory to achieve high mean Average Precision (mAP).⁸

2.2 Multi-Object Tracking (MOT) and Re-Identification

Detection identifies objects in individual frames, but *tracking* links these detections across time to create continuous trajectories. This transforms a set of isolated bounding boxes into a coherent narrative of movement. The primary challenge in football tracking is "identity switching," where the system loses track of a player or confuses two players. This typically occurs during occlusions, such as goal celebrations, corner kicks, or crowded tackles, where players overlap significantly in the camera's field of view.

ByteTrack has emerged as the superior algorithm for this task, significantly outperforming predecessors like DeepSORT in sports contexts.³ DeepSORT relies heavily on visual appearance features (ReID embeddings) to associate detections. While effective in pedestrian

tracking, this approach fails in football because teammates wear identical uniforms, making visual differentiation based on embeddings unreliable.

ByteTrack addresses this by utilizing a hierarchical association method based on detection confidence:

1. **High-Confidence Matching:** The algorithm first associates high-confidence detections with existing tracklets using Intersection over Union (IoU) and motion prediction via Kalman Filters. This handles the "easy" cases where players are clearly visible and moving predictably.
2. **Low-Confidence Recovery:** Crucially, ByteTrack does not discard low-confidence detections, which are often caused by motion blur or partial occlusion. Instead, it attempts to associate these "weak" detections with unmatched tracks. This recovery mechanism is vital for maintaining player IDs during rapid movements or when players are partially obscured by teammates.³

In extended sequences, however, even ByteTrack can suffer from identity switches. To mitigate this, systems employ a secondary "Global Re-Identification" module. This involves extracting visual embeddings for each player and clustering them to ensure that a player who leaves the frame and returns is recognized as the same entity. The achieved Identity F1 Score (IDF1) for such systems typically reaches ~85%, aligning with state-of-the-art benchmarks.³

2.3 Field Registration and Homography Transformation

Raw tracking data exists in "image coordinates" (pixels), which are insufficient for tactical analysis. To derive meaningful metrics like speed (meters per second) or distance covered (kilometers), these coordinates must be converted to "pitch coordinates" (meters). This process, known as field registration or homography estimation, maps the 2D perspective view of the broadcast camera to a standardized 2D top-down view of the pitch (typically 105x68 meters).⁸

The standard approach involves identifying key geometric landmarks on the pitch—corner flags, the center circle, the penalty box lines, and the intersection of field lines. A homography matrix H (a 3×3 transformation matrix) is then computed to map points from the image plane to the ground plane:

$$P_{\text{pitch}} = H \cdot P_{\text{image}}$$

Neural networks, specifically pose-estimation models or semantic segmentation models trained on pitch markings, are used to detect these landmarks automatically.¹² For instance, a model might identify the intersection of the halfway line and the touchline. Once four or more corresponding points are identified between the video frame and the 2D model of the pitch, the homography matrix can be solved.¹³

A significant challenge in broadcast footage is the camera's Pan-Tilt-Zoom (PTZ) movement. As the camera moves, the homography matrix changes. Systems must recalculate H for every frame or use optical flow to track the background movement and update the matrix dynamically.⁴

Visualizing the Transformation:

Consider a standard broadcast frame where players are enclosed in bounding boxes. Through the application of the homography matrix, these pixel coordinates are projected onto a clean, schematic 2D tactical board. Lines connecting specific landmarks in the video—such as the penalty spot or the corner arc—to their corresponding fixed points on the 2D map illustrate the mathematical correspondence utilized to standardize the data.¹¹ This transformation is the bridge between visual data and physical metrics.

3. Semantic Player Analysis: Identity, Role, and Team

Once the system has tracked "objects" on a 2D plane, the next step is to imbue these objects with semantic identity: Which team are they on? What is their jersey number? Are they a goalkeeper? This semantic layer is what turns raw coordinates into a "match."

3.1 Team Assignment via Unsupervised Clustering

Distinguishing between teammates, opponents, and referees relies on color analysis. Since jersey colors vary from match to match (e.g., Home vs. Away kits), supervised classification models trained on specific colors are impractical. Instead, unsupervised clustering algorithms, specifically K-Means, are the standard solution.⁸

The process follows a structured pipeline:

1. **Region of Interest (RoI) Extraction:** For each detected player, the bounding box defining their position is extracted.
2. **Background Removal:** A critical step is to remove the green pitch background from the bounding box to prevent it from skewing the color clustering. This is often achieved using HSV (Hue, Saturation, Value) color histograms to mask out green pixels.¹⁶
3. **K-Means Clustering:** The remaining pixel colors of the player's upper body (the jersey area) are fed into a K-Means clustering algorithm. Typically, $K=2$ is used to separate the two teams. Referees often appear as outliers or can be clustered into a third group if $K=3$ is used.⁴
4. **Temporal Consistency:** Lighting changes or shadows can cause a player's color signature to drift frame-to-frame. To prevent a player from "switching teams" mid-run, a voting mechanism is applied across the entire tracklet. If a player is classified as "Team A" in 90% of frames, the system enforces this label globally for that track ID.¹⁷

3.2 Jersey Number Recognition: The VLM Revolution

Historically, recognizing jersey numbers was one of the most difficult OCR (Optical Character

Recognition) tasks in computer vision. The deformation of shirts due to player movement, the vast variety of fonts used by different leagues, and the low resolution of players in wide broadcast shots made traditional OCR tools like Tesseract highly unreliable.

However, the advent of **Vision-Language Models (VLMs)**, specifically architectures like **Qwen2.5-VL**, has revolutionized this specific sub-task. Qwen2.5-VL is a multimodal model capable of analyzing images with high semantic understanding. Unlike standard OCR, which looks for text patterns, Qwen2.5-VL can be prompted with natural language instructions such as *"Locate the jersey number in this image crop and return it as a JSON integer"*.¹

This approach offers several advantages:

- **Contextual Understanding:** The model understands the concept of a "jersey" and a "number," allowing it to decipher distorted or partially occluded digits that would baffle a standard pattern matcher.
- **Dynamic Resolution:** Qwen2.5-VL supports dynamic resolution processing. This is crucial for football analytics because player crops from wide-angle shots are often small (low resolution). The model can process these small crops without aggressive downscaling, preserving the pixel information necessary to distinguish a '6' from an '8'.¹⁸
- **Legibility Filtering:** In many frames, a player's number is not visible (e.g., they are facing the camera or obscured). A robust system uses the VLM's confidence score or a dedicated "legibility classifier" to identify high-quality frames. The system waits for a clear view of the number, recognizes it once, and then propagates that identity (e.g., "Player #9") to the entire tracklet.¹⁹

3.3 Adaptive Frame Sampling for Efficiency

Running complex models like YOLO, ByteTrack, and especially VLMs like Qwen2.5-VL on every single frame of a 90-minute match is computationally prohibitive and often redundant. To address this, **Adaptive Frame Sampling** or keyframe selection algorithms are employed.²⁰

For tasks like jersey number recognition, the system does not need to run OCR on every frame. Instead, it analyzes the tracklet to find "Smart Stride" frames—moments where the player is upright, facing away from the camera, and motion blur is minimal. The VLM is triggered only on these optimal frames. Once the identity is established with high confidence, it is locked for that trajectory, significantly reducing the computational load.²²

4. Game State Reconstruction: Heuristics and Event Logic

With players tracked and identified, the system acts as a digital scout, "watching" the game to detect discrete events—passes, shots, duels—from the continuous stream of positional data. This process is often referred to as Game State Reconstruction (GSR).¹ It relies on a series of

deterministic decision trees and kinematic heuristics to infer actions from movement.

4.1 Possession Logic

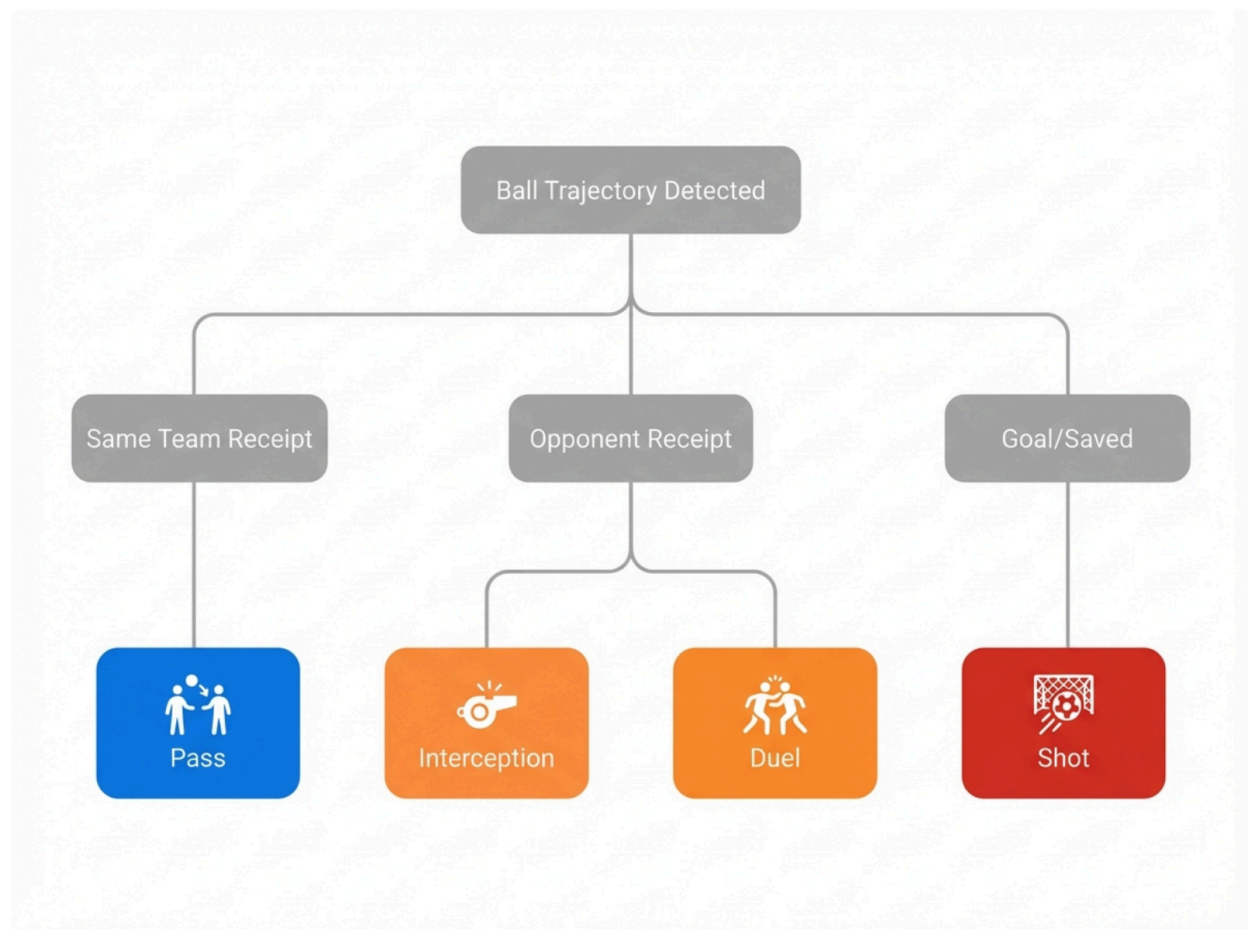
The cornerstone of event detection is determining which player, if any, is in control of the ball. This is typically achieved via a deterministic algorithm based on spatiotemporal thresholds.²

- **Proximity Thresholds:** A player is considered a candidate for possession if the ball is within a defined radius (e.g., 1.5 to 2 meters) of their x, y coordinates. If 3D ball data is available, the z -coordinate is also checked to ensure the ball is near the ground.
- **Dwell Time and Interaction:** Proximity alone is insufficient; a ball passing *near* a player is not possession. The algorithm checks for "Dwell Time" (the ball remaining in the proximity zone for consecutive frames) or a sudden change in the ball's velocity vector, which indicates a kick or touch.²³
- **Possession Chains:** By linking consecutive control frames, the system builds "possession chains." A sequence of touches by players of the same team constitutes a possession phase. A change in the controlling team signals a "Turnover" or "Ball Recovery."

4.2 Classifying Ball Actions: Passes, Shots, and Crosses

Once possession phases are established, the transitions between them—where the ball is moving between players—are classified into event types using rule-based logic.²⁴

Decision Tree: From Trajectories to Match Events



This decision tree illustrates the heuristic logic used to classify on-ball events. The algorithm evaluates the start and end states of the ball's trajectory, the proximity of opponents, and the zones on the pitch (e.g., Penalty Area) to distinguish between Passes, Shots, Crosses, and Interceptions.

- **Pass:** A pass is defined as a ball trajectory starting from Player A (Team X) and ending in the control of Player B (Team X). Algorithms check for a successful reception within a specific time window.
- **Interception:** If a ball trajectory starts from Player A (Team X) but is interrupted or controlled by Player C (Team Y), it is classified as an interception. This requires detecting the moment of possession change without a stoppage in play.
- **Shot:** A shot is a ball trajectory starting from an attacking player that ends in the goal, goes out of play near the goal frame (Shot Wide), or is saved by the goalkeeper. Heuristics such as "direction toward goal" and "velocity > threshold" are critical here. For example, a high-velocity ball moving towards the goal line is flagged as a potential shot.²
- **Cross:** A cross is a specific subset of passes. It is defined spatially: a pass originating from the lateral "flank" zones of the final third and ending in the central penalty area. The

detection logic relies on defining specific polygons on the pitch (zones) to trigger the "Cross" classification.²

4.3 Complex Events: Duels, Tackles, and Set Pieces

Detecting physical contests and dead-ball situations requires more sophisticated heuristics that analyze the interactions between multiple players.

- **Duels and Challenges:** A duel is detected when players from opposing teams are simultaneously within a close radius of the ball (e.g., < 2 meters) and the ball trajectory becomes erratic or changes possession. Aerial duels are inferred if the ball is above a certain height (e.g., $z > 1.8\text{m}$) while two players are in close XY proximity. Success in a duel is attributed to the player who gains control or directs the ball to a teammate.²⁶
- **Tackles:** A tackle event is often inferred when a "Duel" state resolves with a possession change or the ball going out of play. Advanced systems use fractional momentum analysis—if a ball carrier's velocity drops significantly while a defender is close, a tackle or challenge is recorded.²⁸
- **Set Pieces:** Dead-ball situations (corners, free kicks) are detected by analyzing player configurations during "dead time" (when the ball is out of play). For example, a "Corner Kick" is triggered if the ball is static at the corner flag and players are clustered in the box. A "Penalty" is identified by the unique static configuration of the kicker at the penalty spot and the keeper on the line.²

5. Advanced Statistical Metrics

Once the game state has been reconstructed into a structured event log, the system calculates advanced metrics that go beyond simple counting stats. These metrics quantify efficiency, risk, and value, providing the deep insights required by professional analysts.

5.1 Expected Goals (xG)

Definition: Expected Goals (xG) measures the probability of a shot resulting in a goal, ranging from 0 to 1. An xG of 0.35 implies that, historically, 35% of shots taken from that specific location and under those specific conditions result in a goal.

Calculation Logic:

Modern xG models use machine learning algorithms (typically XGBoost or Logistic Regression) trained on hundreds of thousands of historical shots.³⁰ Key features extracted from the tracking data include:

- **Distance to Goal:** The Euclidean distance from the shot coordinates (x, y) to the center of the goal.
- **Shot Angle:** The visible angle of the goal mouth from the shooter's position. This geometric feature is critical; a shot from the center of the box has a much wider angle than one from the byline.³²

- **Defensive Pressure:** Unlike simple event-based xG, tracking-based xG includes the positions of defenders. The model calculates the number of defenders between the ball and the goal and their proximity to the shooter. This "defensive density" is a major accuracy improvement.³³
- **Shot Type:** Whether the shot was taken with the head or foot (headers typically have lower conversion rates).
- **Play Context:** Whether the shot occurred during open play, a set piece, or a counter-attack.

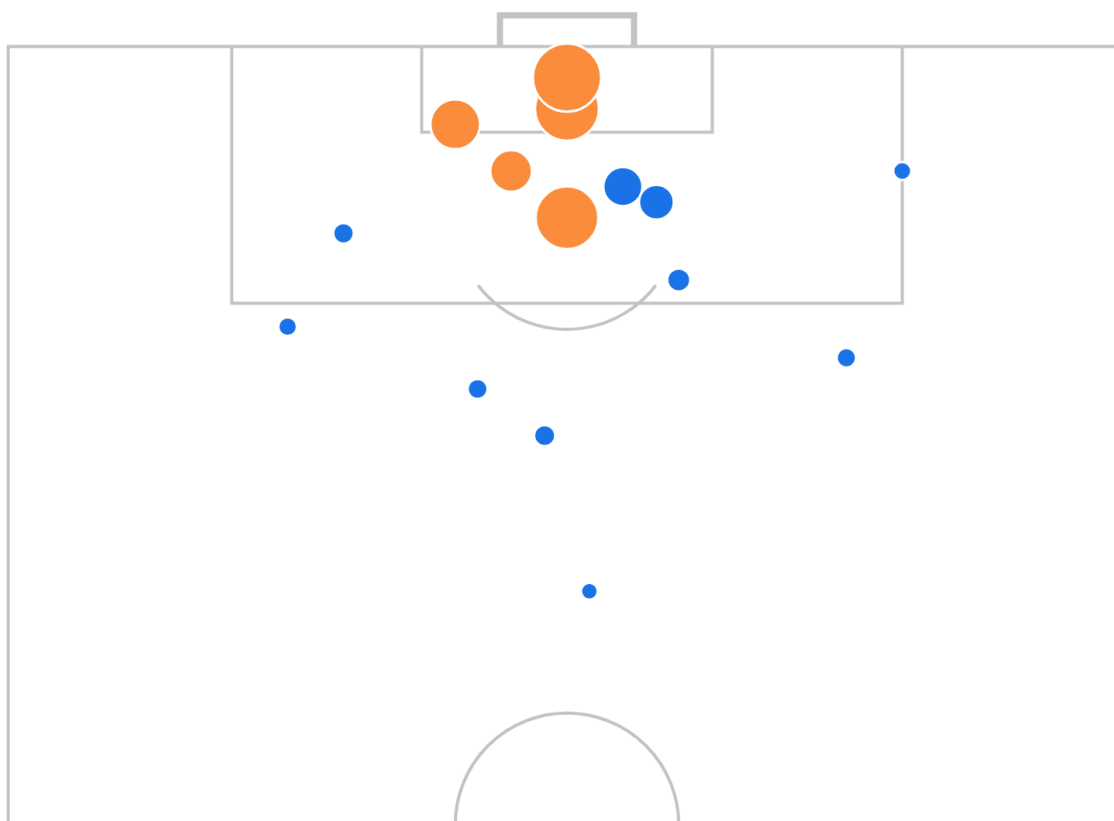
The system tracks metrics such as **xG per shot** (average quality of chances) and **Net xG** (Team xG minus Opponent xG) to evaluate performance beyond the actual scoreline.³⁵

Shot Quality Map: Visualizing Expected Goals (xG)

Attacking Half Performance

Circle size = xG Probability

● Goal ● No Goal (Save/Miss)



This shot map visualizes a player's shooting performance. Each circle represents a shot; the size of the circle corresponds to the xG value (probability of scoring), and the color indicates the outcome (Goal vs. No Goal). High-probability chances closer to the goal appear larger.

Data sources: [Player Stats File](#), [Medium \(xG Model\)](#), [Soccermatics](#)

5.2 Packing and Line-Breaking Passes

Definition: "Packing" is a metric developed to measure the effectiveness of forward passing and dribbling. It counts the number of opponent players "bypassed" or "packed" by an action. The premise is that bypassing defenders is the primary mechanism for creating scoring chances.³⁶

Calculation Logic:

1. **Defensive Line Snapshot:** At the moment a pass is released, the system records the x -coordinates (longitudinal position relative to the goal) of all defending players.
2. **Vector Analysis:** A virtual vertical line is drawn at the x -coordinate of the ball. As the ball travels to the receiver, the algorithm calculates how many defenders were originally *in front* of the ball but are *behind* the ball when it is successfully received.
3. **Weighting:** Not all bypassed players are equal. Bypassing a forward (who does little defending) is less valuable than bypassing a defender. Therefore, packing scores are often weighted: bypassing a defender might be worth 3 points, a midfielder 2, and a forward 1.³⁸

This metric identifies "progressive" players—often deep-lying playmakers or center-backs—who may not register assists but are essential for moving the ball through defensive structures. The analysis file includes Packing (total) as a key performance index.³⁵

5.3 Ball Recovery and Pressing Efficiency

Definition: A Ball Recovery is defined as a player regaining possession for their team when the ball was previously loose or controlled by the opponent.³⁹

Calculation Logic:

The system identifies "Ball Recovery" events by monitoring possession flips in the tracking data.

- **Recovery Time:** A highly tactical metric, "Ball Recovery Time," measures the seconds elapsed between a team losing the ball and regaining it. This serves as a proxy for pressing intensity. Top pressing teams often target a recovery time of under 5 seconds.⁴⁰
- **High Recovery:** Recoveries occurring in the attacking third (the final 35 meters of the pitch) are tagged separately. These "High Recoveries" correlate strongly with creating immediate, high-danger scoring chances due to the opponent's disorganized defensive shape.⁴²

5.4 Expected Assists (xA) and Expected Threat (xT)

While xG values shots, **Expected Assists (xA)** values the creativity of passing. It assigns a probability to a pass becoming an assist based on the likelihood of the *subsequent* shot being scored. If a player makes a pass that leads to a tap-in (0.9 xG shot), the passer receives 0.9 xA, regardless of whether the striker scores or misses. This isolates the creator's quality from the finisher's performance.⁴³

Expected Threat (xT) creates a more holistic value model. It evaluates every action (pass or dribble), not just those leading to shots. The pitch is divided into a grid, where each zone has a pre-calculated "threat value" based on the historical probability of a goal originating from there. Moving the ball from a low-value zone (e.g., defensive wing) to a high-value zone (e.g.,

central area just outside the box) generates positive xT. This rewards players who contribute to the buildup phase, often "the pass before the assist".⁴⁴

6. Implementation Challenges and Solutions

Building a fully automated pipeline presents significant engineering hurdles. The following solutions address common failure modes in computer vision-based sports analytics.

6.1 Occlusion Handling

In crowded penalty areas, players frequently block the camera's view of the ball or each other.

- **Solution:** Kalman Filters (integrated into ByteTrack) predict the trajectory of a player during occlusion. If a player disappears behind another, the filter continues to update their predicted position based on their last known velocity vector. When the player reappears, the tracker re-associates the ID based on the predicted vs. actual position, preventing ID switching.⁹

6.2 The 2D-to-3D Problem

Standard broadcast footage provides only a 2D plane, but football is a 3D game. Knowing the ball's height (z-coordinate) is crucial for distinguishing a ground pass from a cross or chip.

- **Solution:** Monocular 3D estimation models use cues such as the ball's changing size (scaling) and its shadow projection to estimate depth and height. While less accurate than multi-camera systems, heuristic models can infer height: if the ball's size decreases while its y-coordinate (height on screen) increases, it is likely rising in the air.⁴⁶

6.3 Processing Latency

Running YOLO, ByteTrack, Pose Estimation, and VLMs simultaneously is computationally expensive.

- **Solution: Adaptive Frame Sampling** is employed to optimize resource usage. Instead of processing every frame, the system selects "keyframes" where significant motion or events occur. For jersey number recognition—a heavy task—the system waits for a "high-quality" frame where the player is facing the camera and motion blur is minimal, runs the VLM once, and then propagates that identity to the rest of the tracklet.²¹

7. Future Directions: The Age of Large Multimodal Models

The integration of models like **Qwen2.5-VL** signals a move away from specialized, brittle heuristics toward "Generalist Agents." Instead of hard-coding rules for "what is a tackle," future systems will likely feed video clips directly to a Multimodal LLM and ask natural

language questions: *"Did a tackle occur, and was it successful?"*. These models can reason about intent, context, and nuance in a way that mathematical distance thresholds cannot, potentially solving the "subjectivity" problem in football data.⁴⁹

Furthermore, the democratization of this technology means that granular tracking data—once the exclusive domain of elite clubs paying for optical tracking installations—is becoming accessible to semi-pro and amateur teams using nothing more than a standard smartphone camera and cloud-based AI processing. This shift promises to fundamentally alter the landscape of scouting, training, and fan engagement.

Conclusion

The AI-driven analysis of football matches is a composite science, amalgamating computer vision, geometry, and statistical probability. By chaining object detectors (YOLO), robust trackers (ByteTrack), and semantic reasoners (Qwen2.5-VL), we can reconstruct the chaotic reality of a football match into an ordered, queryable database. This data feeds into sophisticated metrics like xG and Packing, providing an objective lens through which to evaluate performance, stripping away the biases of human observation and revealing the underlying mathematical truth of the beautiful game.

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